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Lightweight Object Detection for African Penguin Population Monitoring

BACKGROUND RECAP

Advancements in computer vision and machine learning have transformed penguin conservation research:

TWO-STAGE DETECTORS:

Detection → 2 separate stages:

1. **Region Proposal Stage**: Generate candidate object regions

2. **Classification & Refinement Stage**: Classify regions and refine bounding box

Examples: Faster R-CNN *(outdated technology)*

Characteristics:

Traditionally GPU Dependent, slower inference and computationally intensive

SINGLE-STAGE DETECTORS

Detection → 1 unified regression step

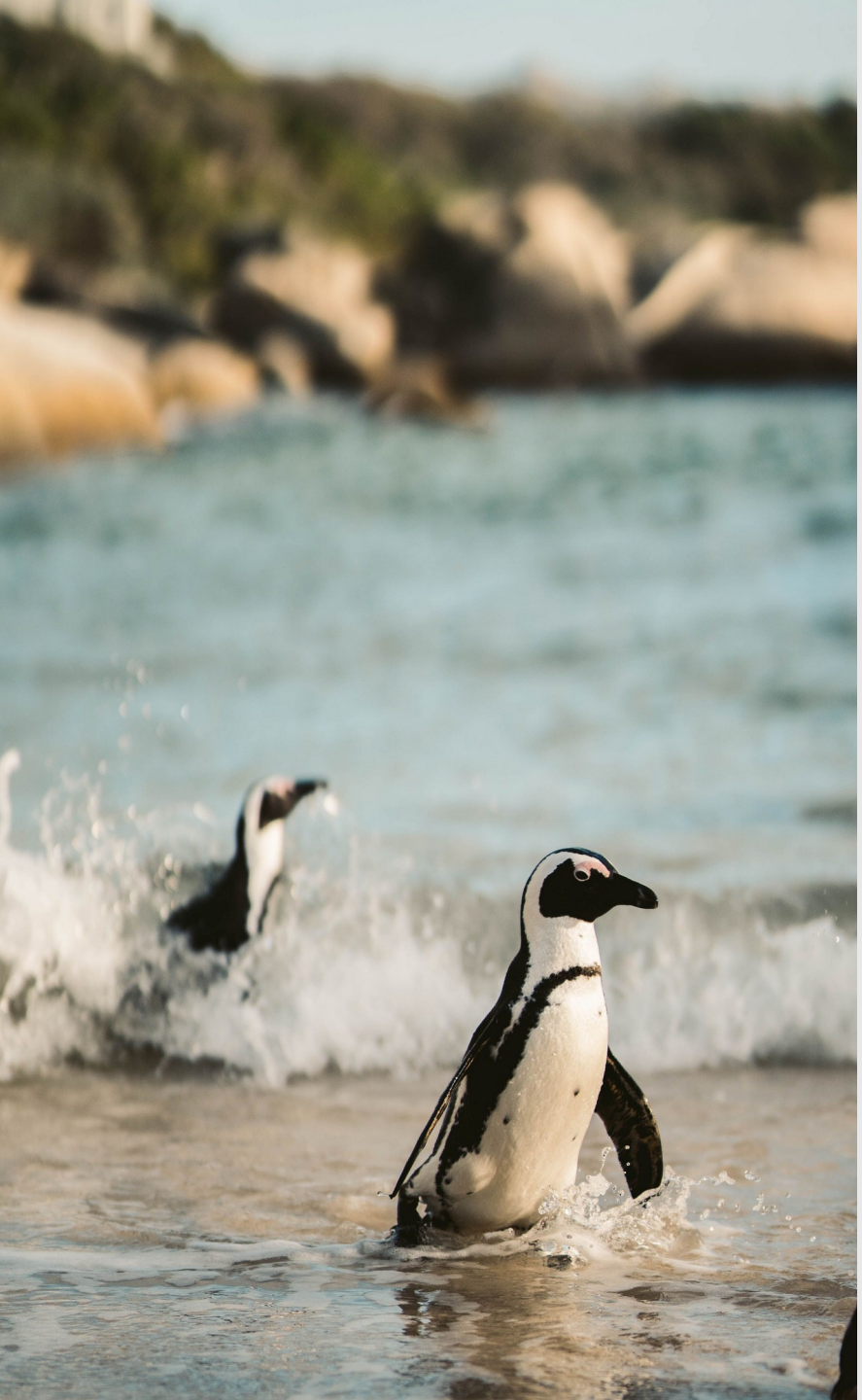
- Image is processed once
- Bounding boxes and class probabilities in the same pass

Examples:
YOLO (You Only Look Once)
YOLOv5n, YOLOv8n
MobileNet-based detectors

Characteristics:

Better aligned with CPU or edge deployment

Mohammed, S. Y. (2025). Architecture review: Two-stage and one-stage object detection. Franklin Open, 100322.

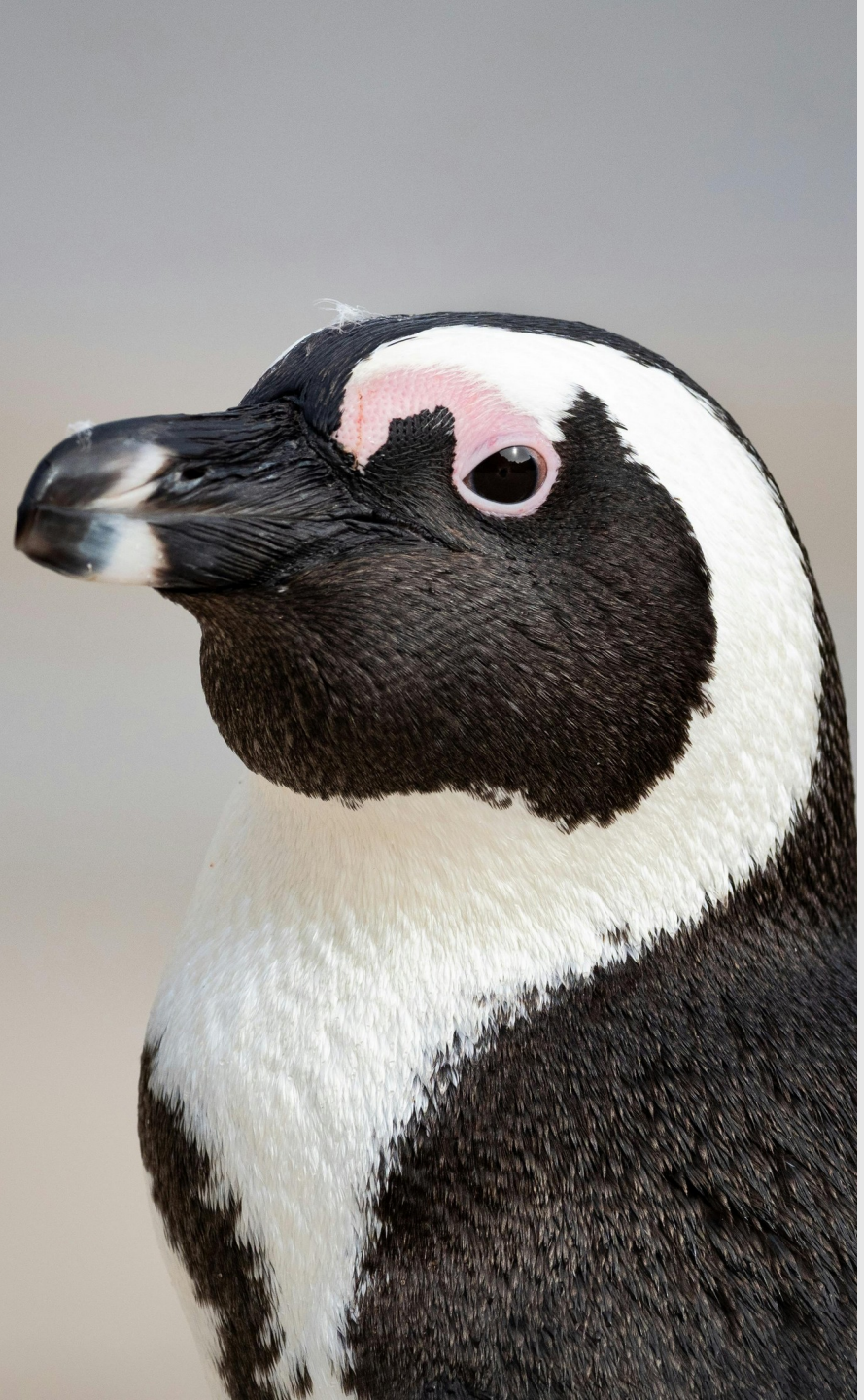


Why care about Lightweight models in the first place?

Conservation monitoring faces practical deployment challenges

- African Penguin populations continue to decline due to climate change and reduced prey availability.
- Effective conservation requires frequent, scalable population monitoring.
- Modern camera systems generate large volumes of imagery, creating a processing bottleneck rather than a data collection bottleneck (Tuia et al., 2022).
- Field researchers often rely on standard CPU laptops or low-power edge devices, where dedicated GPUs are unavailable (Velasco-Montero et al., 2024).

Therefore, conservation requires lightweight object detection models that remain accurate while being deployable on resource-constrained hardware.



African Penguin Detection Studies & Deployment Gap

- **Aerial Imagery Detection**
- *Wu et al. – YOLOPd (Single-Stage)*
- Designed address small-object detection
- +8.5 mAP over Faster R-CNN
- 94.6% counting accuracy
- Evaluated on V100 GPU
- **No CPU deployment analysis**

Adélie Penguin Aerial Dataset
58 aerial images
32,567 annotated penguins
40 train, 18 validation
Images tiled into **3,545 sub-images**

African Penguin Detection Studies & Deployment Gap

- **Aerial Imagery Detection**
- *Liu et al. – Faster R-CNN (Two-Stage)*
- UAV orthomosaic imagery
- Result within 2% of expert counts
- Required GTX-1070 / GTX-2080 GPUs
- **No CPU efficiency evaluation**

Danger Island (2015) + Low Island (2020) UAV Orthomosaic Dataset
4,024 training images
103,641 annotated penguin nests



Problem Statement

African Penguin *(Spheniscus demersus)*

- Endemic species to coastline of Namibia and South Africa
- +/- 73% population decline → IUCN Red List of Threatened/ Endangered Species

Problem:

•Field surveys typically conducted on laptop systems without dedicated GPUs and limited connectivity

•Makes real-time deployment of high-capacity models impractical

•Trade-off between efficiency gains and census-level accuracy remains insufficiently explored under CPU-only, hardware-constrained field conditions.





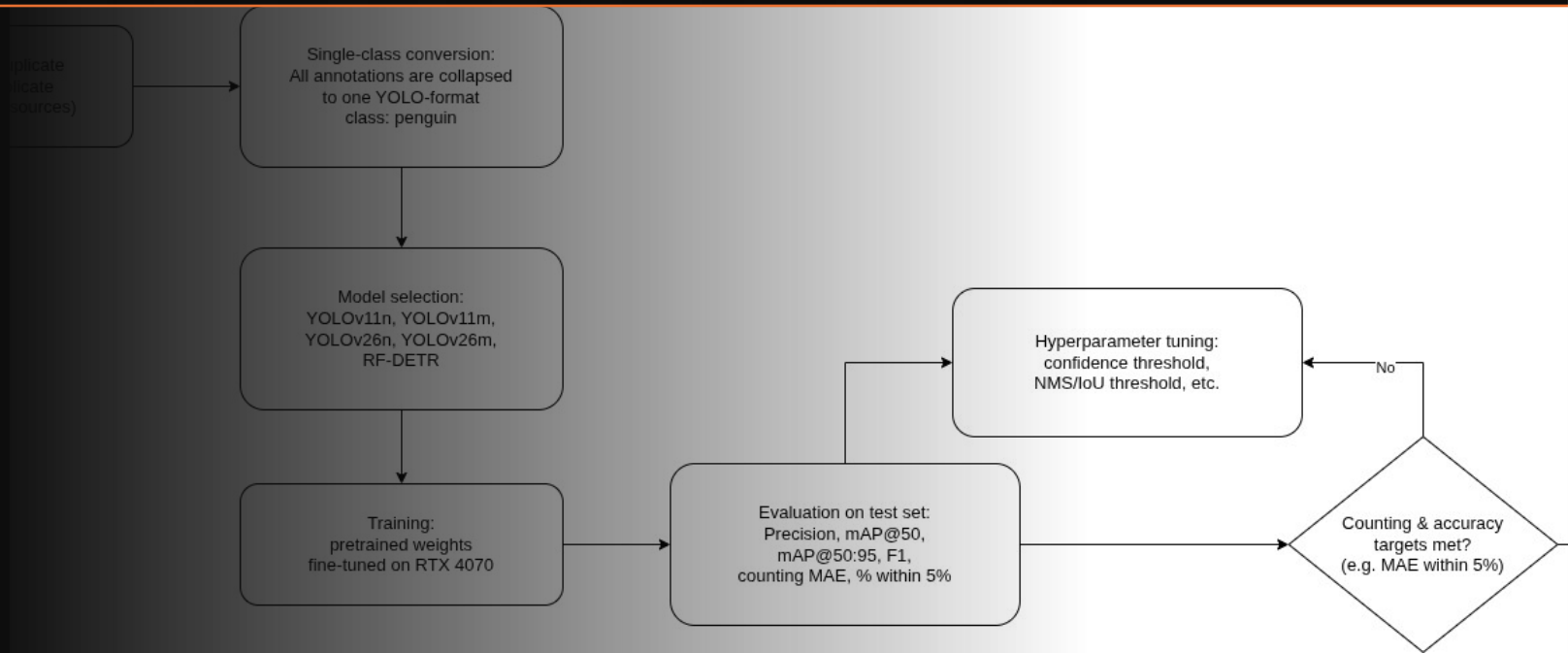
Research Objectives

•RO1: Prepare datasets and develop a lightweight object detection model to measure detection accuracy in real-world conditions.

•RO2: Evaluate inference latency and computational efficiency under CPU and integrated GPU deployment.

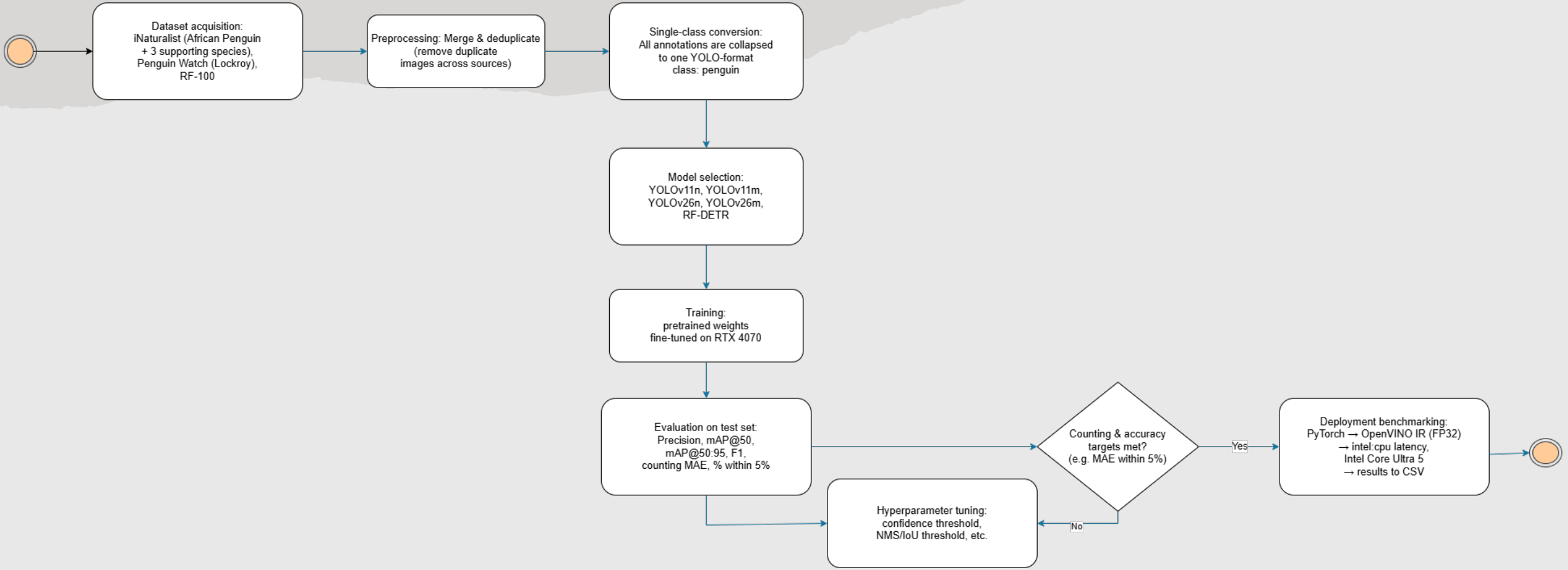
•RO3: Analyse accuracy–efficiency trade-offs between lightweight and higher-capacity models under CPU constraints.

PROPOSED MODEL DESIGN OVERVIEW



PROPOSED MODEL DESIGN OVERVIEW / PIPELINE

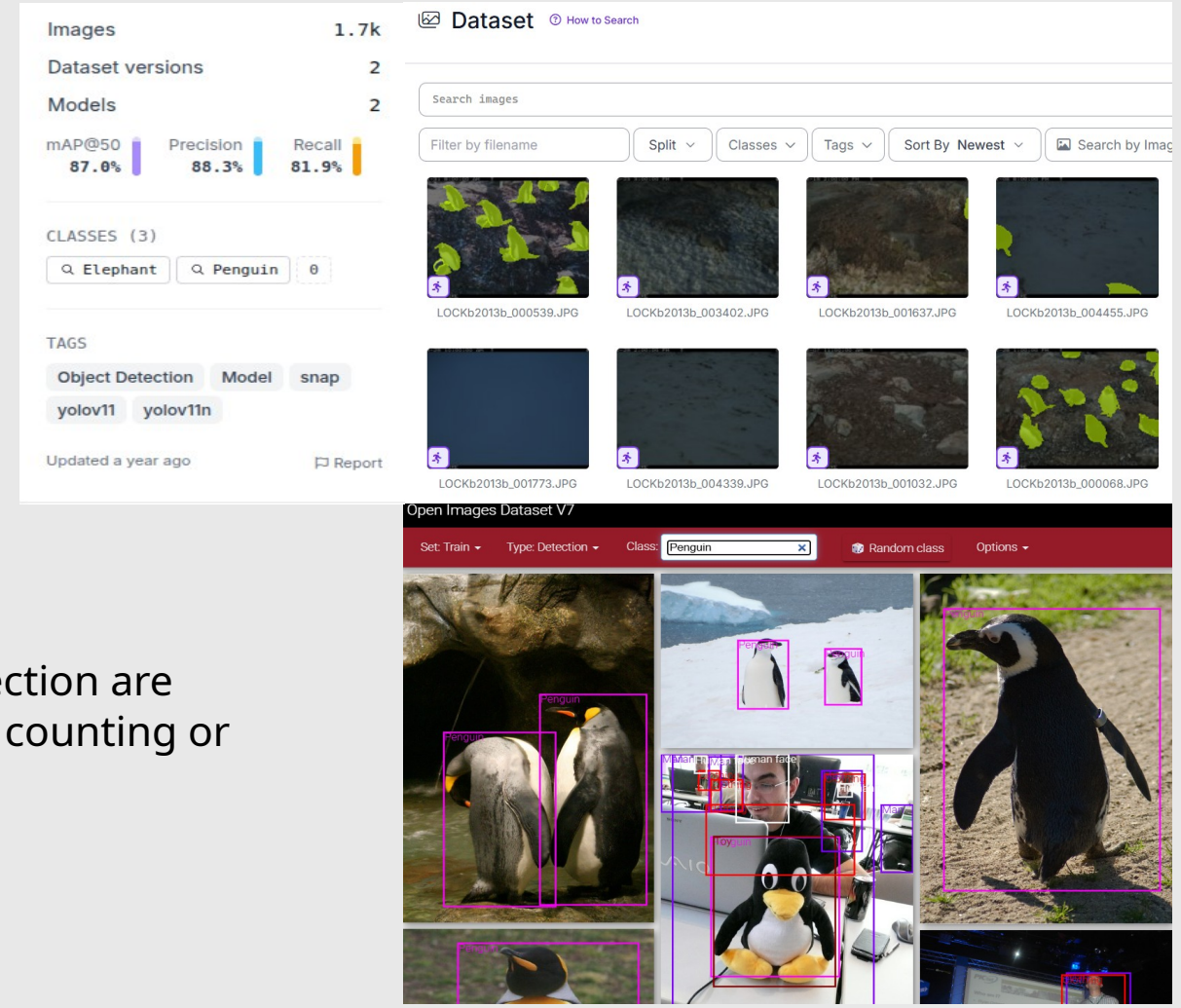
→ RO1



- **High-level overview:** Dataset acquisition → Preprocessing → Model selection → Training → Evaluation →Deployment benchmarkin
- A hyperparameter-tuning loop sits between evaluation and deployment — models are re-evaluated until counting and accuracy targets are met

Dataset Sources

Source	Images used for Training
Roboflow Universe (RF100-VL, Google OpenImagesV7)	Additional imagery
Penguin Watch (Lockroy colony) [1]	5,298
Additional Imagery (Manual Collection and Annotation)	28 900



Existing benchmark datasets for African penguin object detection are limited. Most previous penguin datasets were developed for counting or pose estimation rather than bounding-box object detection.

Annotation Pipeline

Filtered iNaturalist and OIV7 images → Segment Anything Model (SAM3) → Segmentation masks → Bounding boxes generated → YOLO-format labels

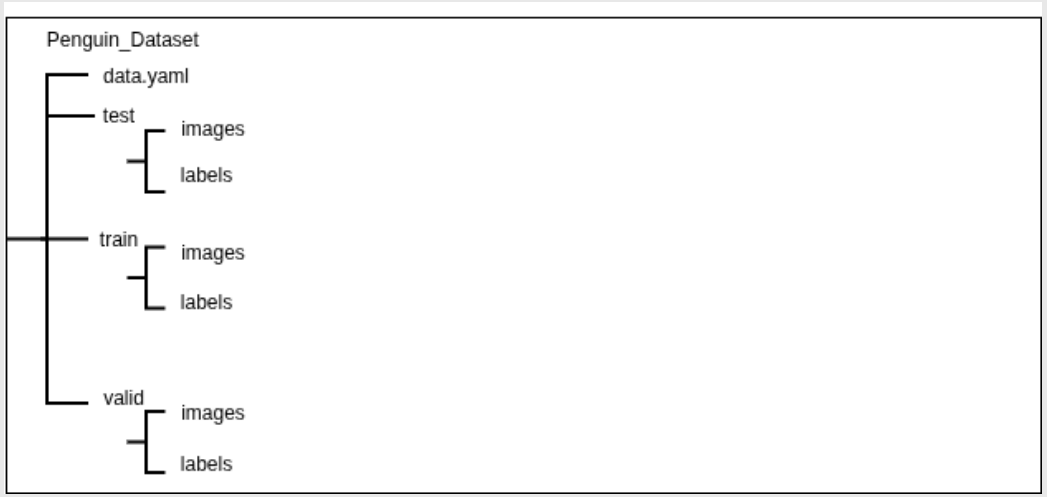
[1] Arteta, C., Lempitsky, V., & Zisserman, A. (2016). "Counting in the Wild." ECCV 2016, LNCS 9911, pp. 483–498.

Preprocessing

Class: Penguin

- All sources merged into one YOLO-format dataset via a preprocessing script
- All annotations collapsed to a single class**: penguin** (detection, not species classification, is the goal)
- Duplicate images removed by hashing image content and discarding exact matches

Structure of Dataset



Final merged dataset structure (YOLO format)

Model Selection & Architecture Overview

→ RO2, RO3

Model	Family	Role
YOLOv11n	YOLO	Primary lightweight detector — edge deployment
YOLOv11m	YOLO	Accuracy ceiling within the v11 family
YOLOv26n	YOLO	Lightweight detector — improved small/dense-object handling
YOLOv26m	YOLO	Accuracy ceiling within the v26 family
RF-DETR	Transformer	Modern SOTA transformer-based comparison point

WHY NANO FOR EDGE DEPLOYMENT?

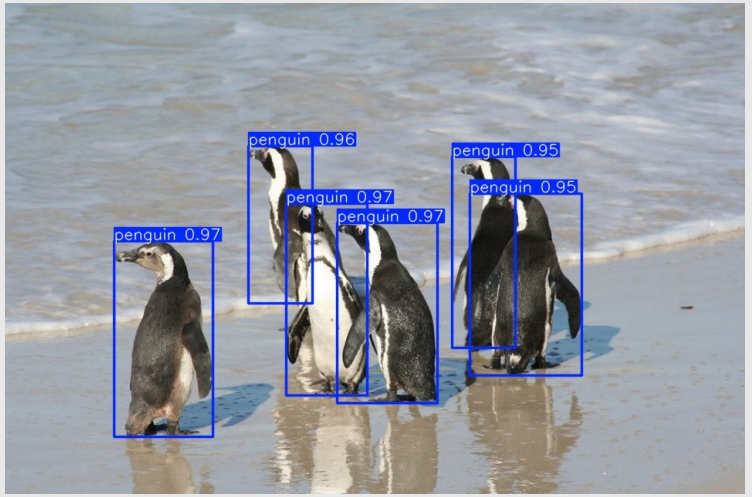
- Deployment constraint: CPU-only field laptops, no dedicated GPU
- Nano variants prioritised for real-time inference under this constraint
- FPS was compared between nano and medium models to measure the speed–accuracy trade-off

Model training

Inference Comparison



Nest scene



Beach scene: 6 Penguins Detected



Occlusion Problem vs Manual Counting

Confidence comparison

Scene	v11m	v26m
Beach (5 penguins)	0.94–0.96	0.95–0.97
Nest-box (occluded scene)	1 detected (0.91)	2 detected (0.94, 0.39)

Playing around with confidence parameters
e.g. conf = 0.25, 0.5, 0.8 doesn't yield differing results

YOLOv26m: consistently higher confidence, and recovers a partially hidden penguin that YOLOv11m missed entirely.

→ RO1, RO3

Preliminary Counting Comparison

- Confidence thresholds were tested from 0.10 to 0.80 on YOLOv26m
- Genuine penguin detections' confidence scores did NOT change across thresholds
- A detection scored 0.94 stays 0.94 regardless of where the threshold is set, it does not change the model's conf score

Do Nano Models Keep Their Advantage for Counting?

Preliminary counting-accuracy comparison, same test set (855 images)

Model	MAE (penguins)	% images within 5%
YOLOv11n	0.88	70.8%
YOLOv26n	0.95	67.8%
—	—	—
YOLOv26m	0.72	67.6%

PRELIMINARY FINDINGS

YOLOv11n achieved a better counting accuracy than YOLOv26n at the **NANO** scale.
Unlike medium models, YOLOv26n did not outperform YOLOv11n.

All models used the **same confidence threshold (0.25)**.
Different thresholds may improve YOLOv26's performance.

Evaluation Metrics & Measures of Success

→ RO1, RO3

Detection

Precision, Recall, F1, mAP@50, mAP@50:95

Counting

Mean Absolute Error (MAE) vs. hand-counted ground truth

Deployment

Inference latency (ms/frame), FPS on Intel CPU

Success target: counting performance within 5% of true counts, consistent with the census-level benchmark from the literature.

Ali, R. et al. (2024); Wu, Y. et al. (2023) — standard detection metrics. Liu, Z. et al. (2020) — 5% census-level counting benchmark.

Why is this Computer Science?

1. Research Gap: Existing penguin detection methods rely on GPU-heavy, high-resolution environment, CPU-only resource-constrained deployment hasn't been addressed to solve conservation efforts

2. Comparison of lightweight and higher-capacity YOLO variants (direction for future exploration: NMS vs NMS-free)

3. Quantifying accuracy-efficiency trade-offs with metrics, mAP, F1, mAE etc.

4. Explores systems optimization: Using inference engines (OpenVINO) and hardware CPU, iGPU, GPUs

5. Contributes to ecological monitoring and CS on efficient ML deployment representative of field deployment



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Timeline

Date	Milestone / Activity
23 Feb 2026	Draft proposal submission
24 Feb – 10 Mar 2026	Seminar Series 1: Proposal presentation and feedback integration
16 Mar 2026	Finalised proposal submission
4 May 2026	Literature Review submission (updated problem statement, questions, objectives)
May 2026	Dataset preparation and bounding box generation script implementation
15 Jun 2026	Begin drafting research design / methodology chapter
VAC (Jun/Jul)	Model implementation and baseline training
7 - 14 Jul 2026	Seminar Series 2: Project implementation presentation
31 Jul 2026	Progress report submission
Aug 2026	OpenVINO optimization and CPU/iGPU benchmarking
Sep 2026	Accuracy–efficiency trade-off analysis and statistical evaluation
18 Sep 2026	Full draft research report submission
2 Oct 2026	Short paper submission (if required)
12–14 Oct 2026	Seminar Series 3: Final results presentation
16 Oct 2026	Final thesis submission
5 Nov 2026	Oral examination
6 Nov 2026	Final corrected submissions (report and ACM short paper)

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Q & A